WORKED SOLUTION

STEP II 1990, Question 2

Pure • Ptolemy's Theorem

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q2

The key idea

The first part is a trigonometric identity that falls apart once every product of sines is turned into a difference of cosines; the condition $A + B + C + D = \pi$ is then used to rewrite $\cos(A + C)$ and $\cos(A - C)$. The second part is Ptolemy's theorem. The centre O is the hint: every chord of a circle of radius r has length $2r\sin(\text{half the angle it subtends at } O)$, and the four half-angles around O add up to π, which is exactly the condition in the first part.

Solution

The identity

We use the product-to-sum formula

$$\sin X \sin Y = \tfrac{1}{2}[\cos(X - Y) - \cos(X + Y)]$$

three times. On the left-hand side,

$$\sin(A + B)\sin(A + D) = \tfrac{1}{2}[\cos(B - D) - \cos(2A + B + D)],$$
$$\sin B \sin D = \tfrac{1}{2}[\cos(B - D) - \cos(B + D)],$$

so the $\cos(B - D)$ terms cancel and

$$\sin(A + B)\sin(A + D) - \sin B \sin D = \tfrac{1}{2}[\cos(B + D) - \cos(2A + B + D)].$$

On the right-hand side,

$$\sin A \sin C = \tfrac{1}{2}[\cos(A - C) - \cos(A + C)].$$

Now we use $C = \pi - A - B - D$. This gives

$$A + C = \pi - (B + D), \qquad \text{so} \quad \cos(A + C) = -\cos(B + D),$$
$$A - C = (2A + B + D) - \pi, \qquad \text{so} \quad \cos(A - C) = -\cos(2A + B + D),$$

using $\cos(\pi-\theta) = -\cos\theta$ and $\cos(\theta-\pi) = -\cos\theta$. Therefore

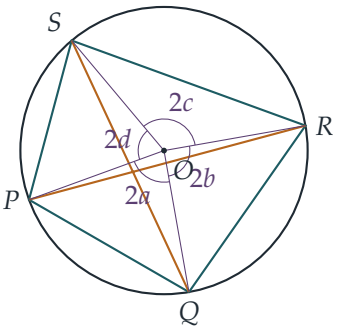
$$\sin A \sin C = \tfrac{1}{2}[-\cos(2A+B+D) + \cos(B+D)],$$

which is the same as the left-hand side. This proves the identity.

The circle

Let the circle have radius r. Going round the circle from P to Q to R to S and back to P, let the arcs PQ, QR, RS, SP subtend angles $2a$, $2b$, $2c$, $2d$ at O. Since the points are in that order, these four arcs make up the whole circle without overlapping, so

$$a, b, c, d > 0, \qquad 2a+2b+2c+2d = 2\pi, \qquad \text{i.e.} \quad a+b+c+d = \pi.$$



Ask yourself How long is a chord whose arc subtends an angle 2θ at the centre?

If $0 < 2\theta \leq \pi$, the triangle formed by O and the two ends of the chord is isosceles with two sides r and apex angle 2θ. The perpendicular from O bisects both the chord and the apex angle, so the chord has length $2r\sin\theta$. If instead $\pi < 2\theta < 2\pi$, the angle at O inside the triangle is $2\pi - 2\theta$, and the chord has length $2r\sin(\pi-\theta) = 2r\sin\theta$. So in every case

$$\text{chord} = 2r\sin\theta.$$

Applying this to the four sides, and to the diagonals (the arc $P \to Q \to R$ subtends $2(a+b)$ and the arc $Q \to R \to S$ subtends $2(b+c)$), we get

$$PQ = 2r\sin a, \quad QR = 2r\sin b, \quad RS = 2r\sin c, \quad PS = 2r\sin d,$$

$$PR = 2r\sin(a+b), \qquad QS = 2r\sin(b+c).$$

Note that $a+b$ or $b+c$ may exceed $\frac{\pi}{2}$; this is why we needed the formula for arcs of more than a semicircle.

After dividing by $4r^2$, the result we want becomes

$$\sin a \sin c + \sin b \sin d = \sin(a+b)\sin(b+c),$$

or, equivalently,

$$\sin(b+a)\sin(b+c) - \sin a \sin c = \sin b \sin d.$$

This is exactly the identity from the first part with $A = b$, $B = a$, $D = c$ and $C = d$, which is allowed because $b + a + d + c = \pi$. Multiplying back by $4r^2$,

$$PQ \times RS + QR \times PS = PR \times QS,$$

as required.

Checks

- A square inscribed in a circle of radius $\frac{1}{\sqrt{2}}$ has sides 1 and diagonals $\sqrt{2}$: $1 \cdot 1 + 1 \cdot 1 = 2 = \sqrt{2} \cdot \sqrt{2}$.
- For any rectangle with sides p and q, Ptolemy's theorem reads $p^2 + q^2 = (\text{diagonal})^2$: Pythagoras's theorem is a special case. For a 3×4 rectangle, $9 + 16 = 25 = 5 \times 5$.
- A numerical check of the identity at 10^5 random values of A, B, D (with $C = \pi - A - B - D$) gave a largest error of about 5×10^{-15}, and 10^5 random cyclic quadrilaterals satisfied the final result to within a relative error of 10^{-15}. With the points taken out of order (for example P, R, Q, S in order round the circle) the equality fails, so the ordering really matters.

Common slips

- Using inscribed angles at a point on the circle without saying which arc they stand on. Working with angles at the centre, as the question suggests, avoids this.
- Forgetting that a diagonal can cut off an arc of more than a semicircle; the formula $2r \sin\theta$ still holds, but it needs the one-line justification above.
- Matching the wrong angles to A, B, C, D: A and C must be the two angles that appear only on the right-hand side, here b and d.

Beyond STEP

Ptolemy's theorem has a far-reaching sequel: for *any* four points in the plane, $PQ \cdot RS + QR \cdot PS \geq PR \cdot QS$, with equality exactly when the points lie in order on a circle (or a line). A slick proof treats the points as complex numbers p, q, r, s, notes the algebraic identity $(p - q)(r - s) + (q - r)(p - s) = (p - r)(q - s)$, and applies the triangle inequality; the equality case is where the geometry of the complex plane comes in. Needham's *Visual Complex Analysis* is a beautiful place to see this kind of argument, and inversion, which also proves the result neatly.

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